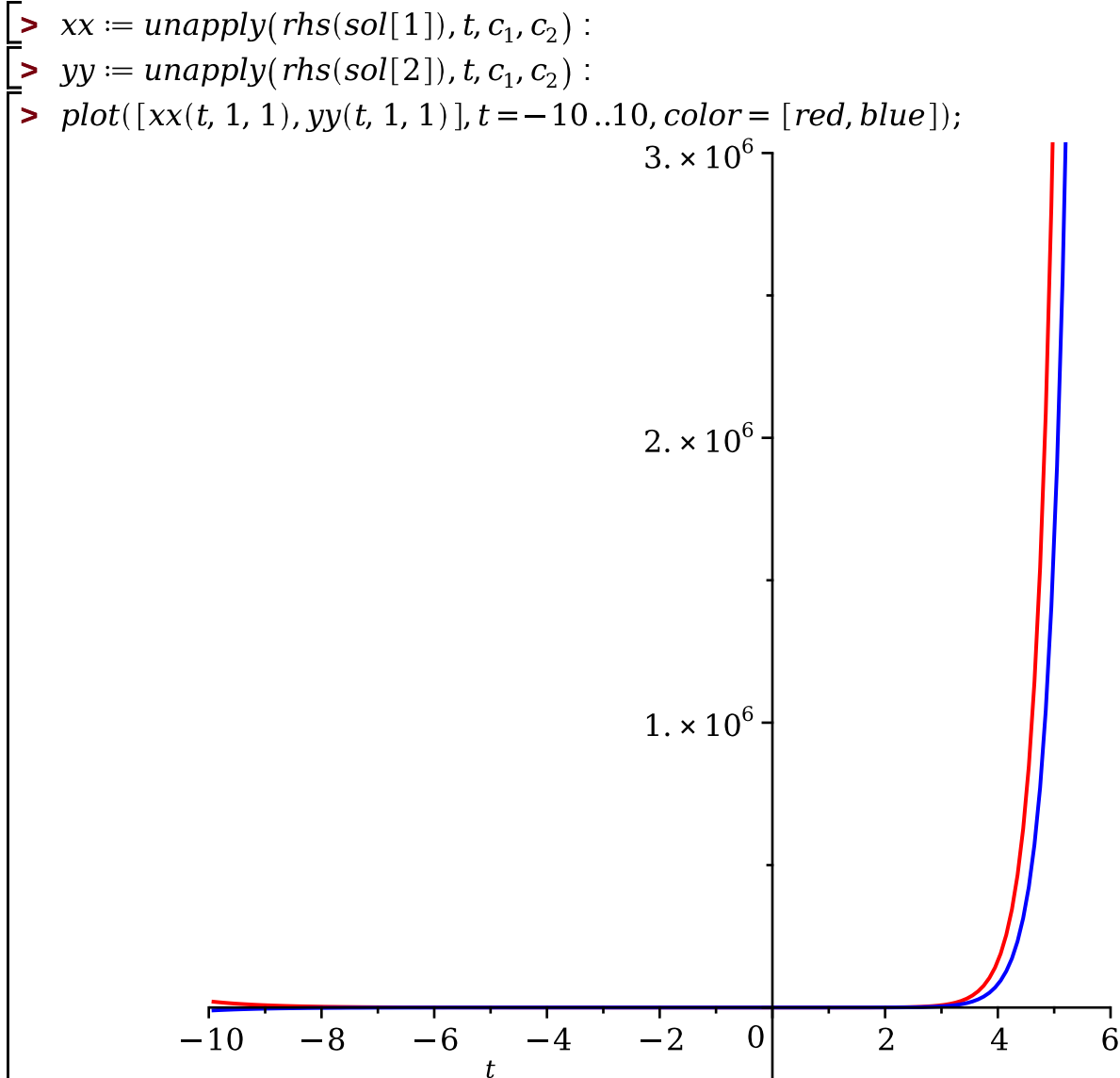


```

> with(DETools):
  ecdif1 := diff(x(t), t) = x(t) + 4·y(t):
> ecdif2 := diff(y(t), t) = x(t) + y(t):
> sist := ecdif1, ecdif2:
> sol := dsolve({sist}, {x(t), y(t)});
      
$$\text{sol} := \left\{ x(t) = c_1 e^{3t} + c_2 e^{-t}, y(t) = \frac{c_1 e^{3t}}{2} - \frac{c_2 e^{-t}}{2} \right\} \quad (1)$$


```



```

> restart:
> with(DETools):
> ecdif1 := diff(x(t), t) = 2·x(t) - y(t):
> ecdif2 := diff(y(t), t) = x(t) + 2·y(t):
> sist := ecdif1, ecdif2:
> sol := dsolve({sist}, {x(t), y(t)});
      
$$\text{sol} := \{ x(t) = e^{2t} (c_2 \cos(t) + c_1 \sin(t)), y(t) = -e^{2t} (\cos(t) c_1 - \sin(t) c_2) \} \quad (2)$$

> xx := unapply(rhs(sol[1]), t, c1, c2);

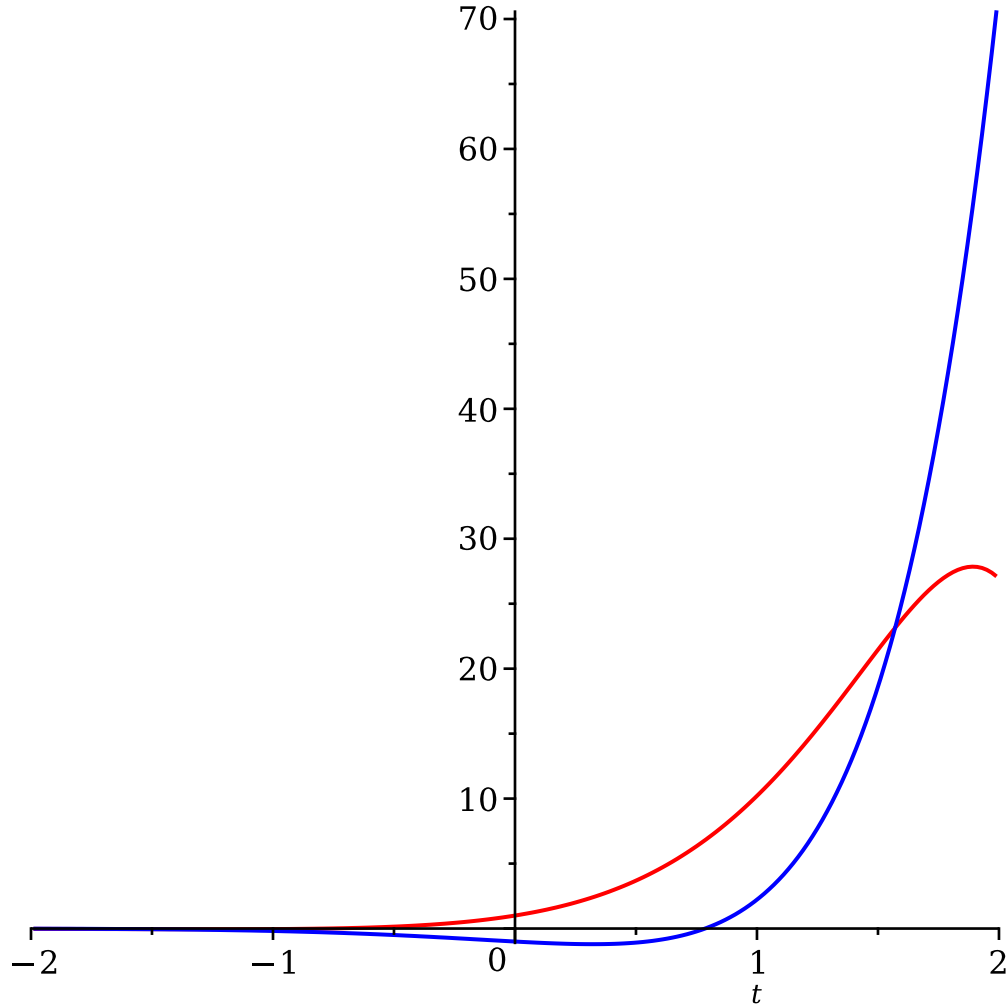
```

$$xx := (t, c_1, c_2) \mapsto e^{2 \cdot t} \cdot (c_2 \cdot \cos(t) + c_1 \cdot \sin(t)) \quad (3)$$

```
> yy := unapply(rhs(sol[2]), t, c_1, c_2);
```

$$yy := (t, c_1, c_2) \mapsto -e^{2 \cdot t} \cdot (\cos(t) \cdot c_1 - \sin(t) \cdot c_2) \quad (4)$$

```
> plot([xx(t, 1, 1), yy(t, 1, 1)], t = -2..2, color = [red, blue]);
```



```
> restart:
```

```
> with(DETools):
```

```
> ecdif1 := diff(x(t), t) = x(t) + 4*y(t):
```

```
> ecdif2 := diff(y(t), t) = x(t) + y(t):
```

```
> cond_in := x(0) = 1, y(0) = 2:
```

```
> sist := ecdif1, ecdif2:
```

```
> sol := dsolve({sist, cond_in}, {x(t), y(t)});
```

$$sol := \left\{ x(t) = \frac{5e^{3t}}{2} - \frac{3e^{-t}}{2}, y(t) = \frac{5e^{3t}}{4} + \frac{3e^{-t}}{4} \right\} \quad (5)$$

```
> xx := unapply(rhs(sol[1]), t);
```

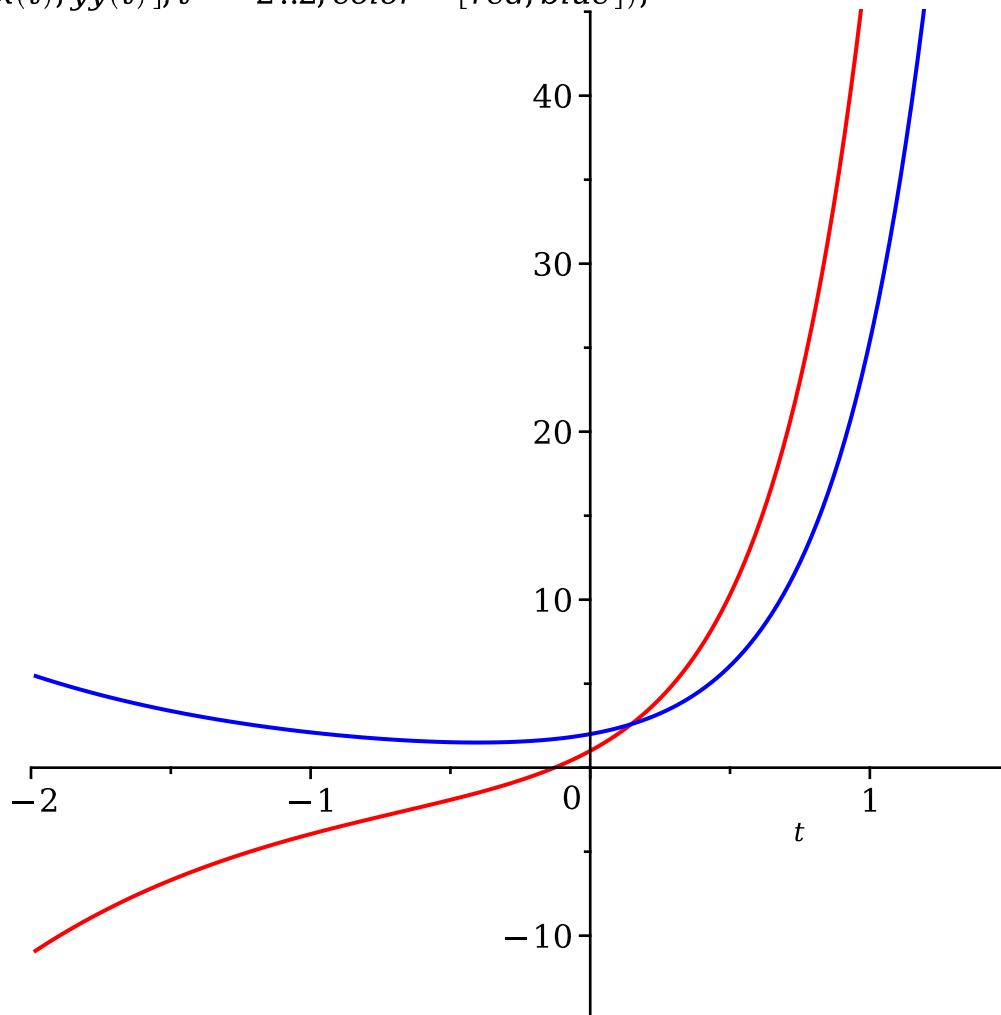
$$xx := t \mapsto \frac{5 \cdot e^{3 \cdot t}}{2} - \frac{3 \cdot e^{-t}}{2} \quad (6)$$

```
> yy := unapply(rhs(sol[2]), t);
```

$$yy := t \mapsto \frac{5 \cdot e^{3 \cdot t}}{4} + \frac{3 \cdot e^{-t}}{4}$$

(7)

```
> plot([xx(t), yy(t)], t = -2..2, color = [red, blue]);
```



```
> restart:
```

```
> with(DETools):
```

```
> ecdif1 := diff(x(t), t) = x(t) - y(t) + t - 1:
```

```
> ecdif2 := diff(y(t), t) = -2*x(t) + y(t) + 1:
```

```
> cond_in := x(0) = 0, y(0) = 1:
```

```
> sist := ecdif1, ecdif2:
```

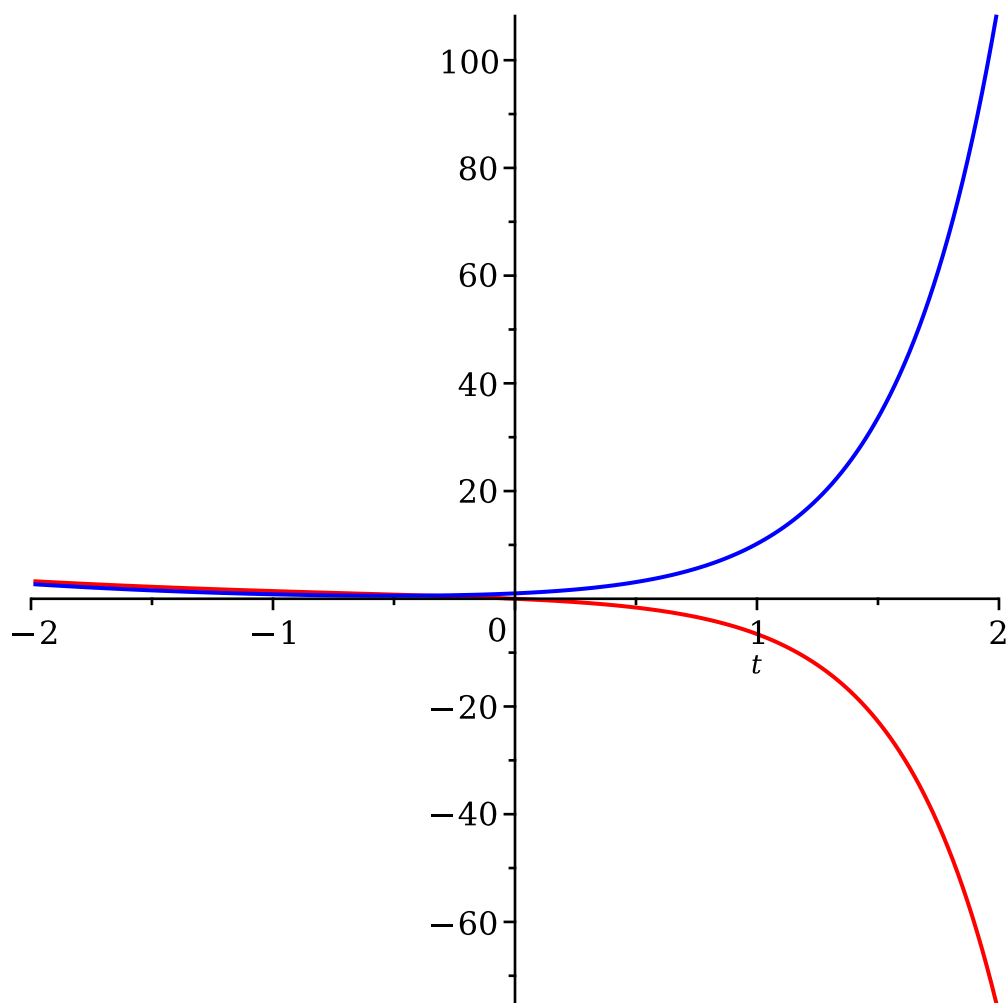
```
> sol := dsolve({sist, cond_in}, {x(t), y(t)});
```

$$sol := \left\{ x(t) = e^{(1+\sqrt{2})t} \left(\frac{3}{2} - \frac{3\sqrt{2}}{2} \right) + e^{-(\sqrt{2}-1)t} \left(\frac{3}{2} + \frac{3\sqrt{2}}{2} \right) + t - 3, y(t) = \right. \\ \left. -e^{(1+\sqrt{2})t} \left(\frac{3}{2} - \frac{3\sqrt{2}}{2} \right) \sqrt{2} + e^{-(\sqrt{2}-1)t} \left(\frac{3}{2} + \frac{3\sqrt{2}}{2} \right) \sqrt{2} + 2t - 5 \right\} \quad (8)$$

```
> xx := unapply(rhs(sol[1]), t):
```

```
> yy := unapply(rhs(sol[2]), t):
```

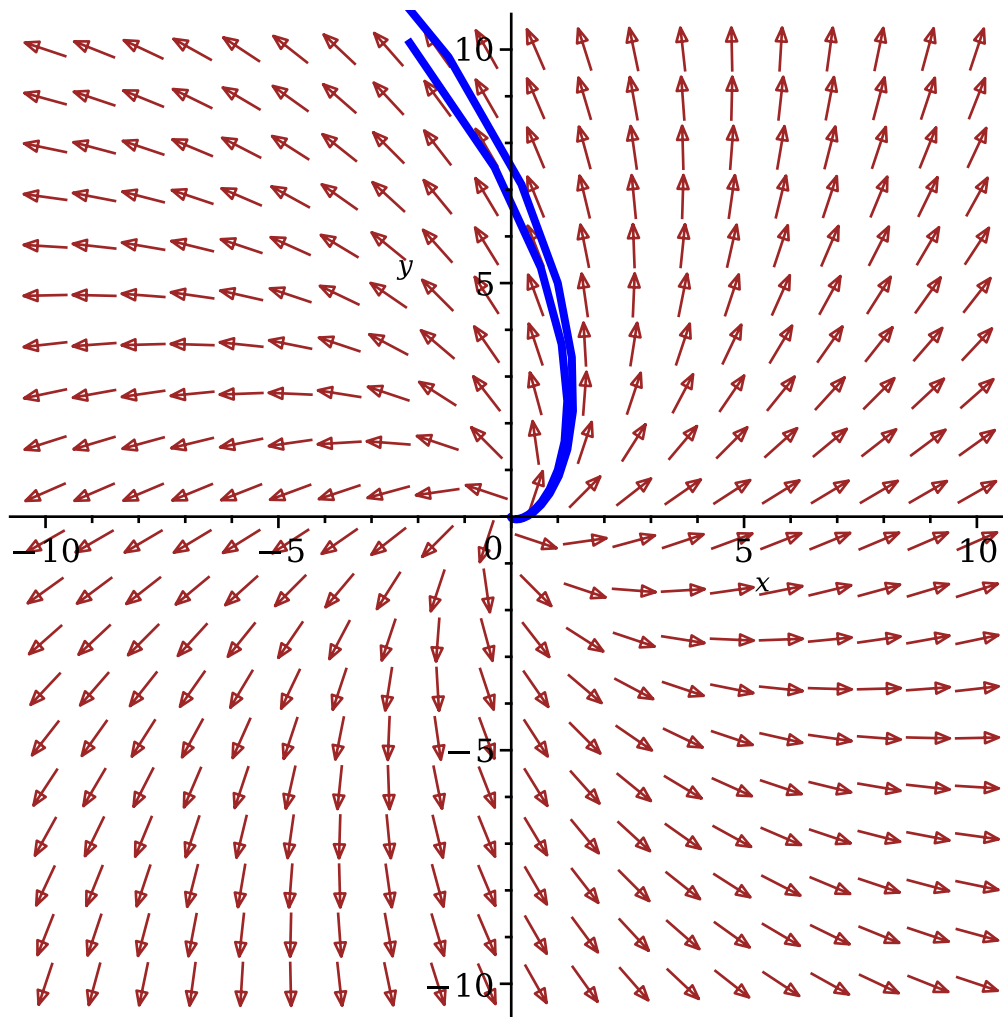
```
> plot([xx(t), yy(t)], t = -2..2, color = [red, blue]);
```



```

> restart :
> with(DETools) :
> ecdif1 := diff(x(t), t) = y(t) :
> ecdif2 := diff(y(t), t) = -x(t) - 2·y(t) :
> sist := ecdif1, ecdif2 :
> sol := dsolve({sist}, {x(t), y(t)});
      sol := {x(t) = e-t (c2 t + c1), y(t) = -e-t (c2 t + c1 - c2) }      (9)
> xx := unapply(rhs(sol[1]), t, c1, c2) :
> yy := unapply(rhs(sol[2]), t, c1, c2) :
> eval(limit(xx(t, c1, c2), t = infinity));
      0      (10)
> eval(limit(yy(t, c1, c2), t = infinity));
      0      (11)
> DEplot([sist], [x(t), y(t)], t = -4..4, x = -10..10, y = -10..10, [[x(0) = 1, y(0)
    = 1], [x(0) = 1, y(0) = 5]], arrows = medium, linecolor = blue);

```



```

> restart :
> with(DETools) :
> ecdif1 := diff(x(t), t) = 2·x(t) + y(t) :
> ecdif2 := diff(y(t), t) = x(t) + 2·y(t) :
> sist := ecdif1, ecdif2 :
> cond_in2 :=
  [x(0) = 0, y(0) = 1],
  [x(0) = 0, y(0) = 2],
  [x(0) = 0, y(0) = 3],
  [x(0) = 0, y(0) = 4],
  [x(0) = 0, y(0) = 5],
  [x(0) = 0, y(0) = -1],
  [x(0) = 0, y(0) = -2],
  [x(0) = 0, y(0) = -3],
  [x(0) = 0, y(0) = -4],
  [x(0) = 0, y(0) = -5],
  [x(0) = 1, y(0) = 0],
  [x(0) = 2, y(0) = 0],
  [x(0) = 3, y(0) = 0],
  [x(0) = 4, y(0) = 0],

```

```

[x(0) = 5, y(0) = 0],
[x(0) = -1, y(0) = 0],
[x(0) = -2, y(0) = 0],
[x(0) = -3, y(0) = 0],
[x(0) = -4, y(0) = 0],
[x(0) = -5, y(0) = 0]

```

```
;
```

```

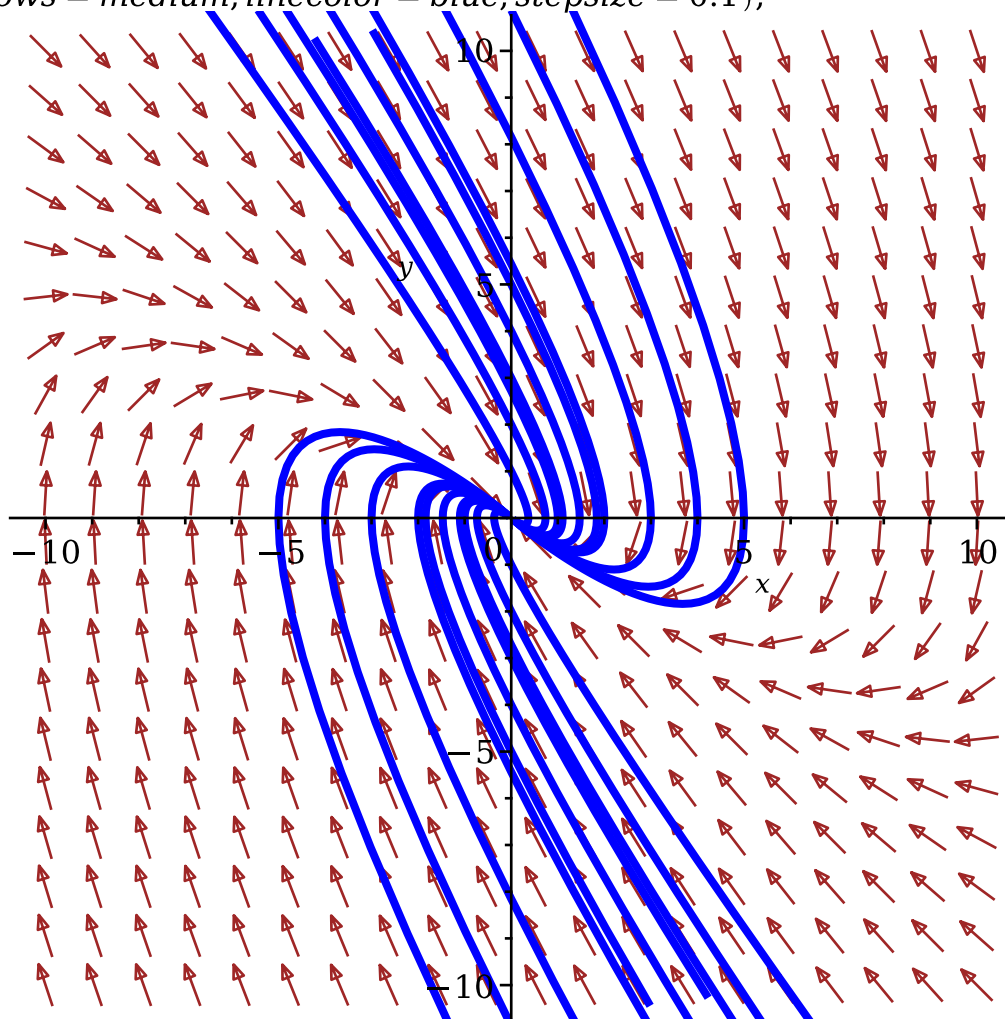
cond_in2 := [x(0) = 0, y(0) = 1], [x(0) = 0, y(0) = 2], [x(0) = 0, y(0) = 3], [x(0) = 0, y(0) = 4], [x(0) = 0, y(0) = 5], [x(0) = 0, y(0) = -1], [x(0) = 0, y(0) = -2], [x(0) = 0, y(0) = -3], [x(0) = 0, y(0) = -4], [x(0) = 0, y(0) = -5], [x(0) = 1, y(0) = 0], [x(0) = 2, y(0) = 0], [x(0) = 3, y(0) = 0], [x(0) = 4, y(0) = 0], [x(0) = 5, y(0) = 0], [x(0) = -1, y(0) = 0], [x(0) = -2, y(0) = 0], [x(0) = -3, y(0) = 0], [x(0) = -4, y(0) = 0], [x(0) = -5, y(0) = 0]

```

```

> DEplot([sist], [x(t), y(t)], t = -5..5, x = -10..10, y = -10..10, [cond_in2],
arrows = medium, linecolor = blue, stepsize = 0.1);

```



```

> restart :
> with(DETools) :
> ecdif1 := diff(x(t), t) = y(t) :
> ecdif2 := diff(y(t), t) = -x(t) :

```

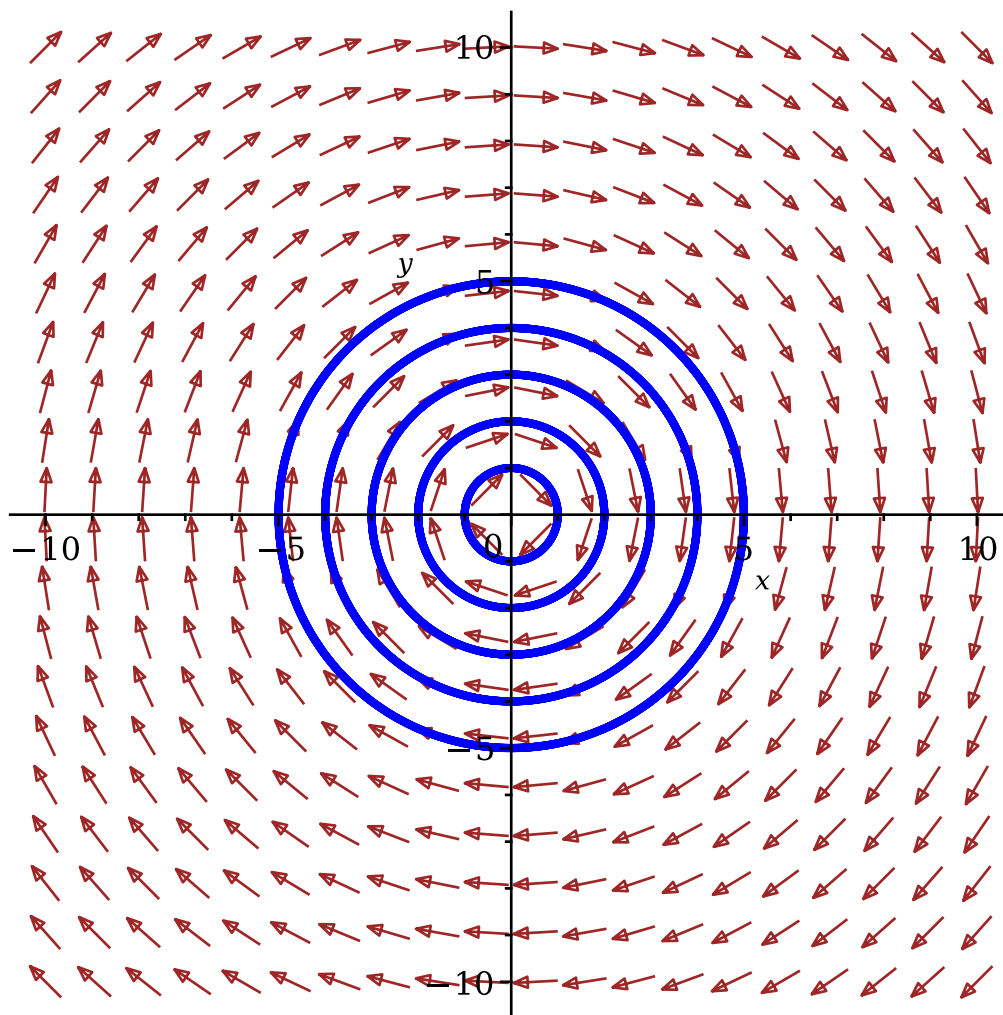
```
> sist := ecdif1, ecdif2 :
```

```
> cond_in2 :=
```

```
[x(0) = 0, y(0) = 1],  
[x(0) = 0, y(0) = 2],  
[x(0) = 0, y(0) = 3],  
[x(0) = 0, y(0) = 4],  
[x(0) = 0, y(0) = 5],  
[x(0) = 0, y(0) = -1],  
[x(0) = 0, y(0) = -2],  
[x(0) = 0, y(0) = -3],  
[x(0) = 0, y(0) = -4],  
[x(0) = 0, y(0) = -5],  
[x(0) = 1, y(0) = 0],  
[x(0) = 2, y(0) = 0],  
[x(0) = 3, y(0) = 0],  
[x(0) = 4, y(0) = 0],  
[x(0) = 5, y(0) = 0],  
[x(0) = -1, y(0) = 0],  
[x(0) = -2, y(0) = 0],  
[x(0) = -3, y(0) = 0],  
[x(0) = -4, y(0) = 0],  
[x(0) = -5, y(0) = 0]
```

```
cond_in2 := [x(0) = 0, y(0) = 1], [x(0) = 0, y(0) = 2], [x(0) = 0, y(0) = 3], [x(0) = 0, y(0) = 4], [x(0) = 0, y(0) = 5], [x(0) = 0, y(0) = -1], [x(0) = 0, y(0) = -2], [x(0) = 0, y(0) = -3], [x(0) = 0, y(0) = -4], [x(0) = 0, y(0) = -5], [x(0) = 1, y(0) = 0], [x(0) = 2, y(0) = 0], [x(0) = 3, y(0) = 0], [x(0) = 4, y(0) = 0], [x(0) = 5, y(0) = 0], [x(0) = -1, y(0) = 0], [x(0) = -2, y(0) = 0], [x(0) = -3, y(0) = 0], [x(0) = -4, y(0) = 0], [x(0) = -5, y(0) = 0]
```

```
> DEplot([sist], [x(t), y(t)], t = -5..5, x = -10..10, y = -10..10, [cond_in2],  
arrows = medium, linecolor = blue, stepsize = 0.1);
```



```

> restart:
> with(DETools):
> ecdif1 := diff(x(t), t) = -x(t) - y(t) :
> ecdif2 := diff(y(t), t) = x(t) - y(t) :
> sist := ecdif1, ecdif2 :
> cond_in2 :=
  [x(0) = 0, y(0) = 1],
  [x(0) = 0, y(0) = 2],
  [x(0) = 0, y(0) = 3],
  [x(0) = 0, y(0) = 4],
  [x(0) = 0, y(0) = 5],
  [x(0) = 0, y(0) = -1],
  [x(0) = 0, y(0) = -2],
  [x(0) = 0, y(0) = -3],
  [x(0) = 0, y(0) = -4],
  [x(0) = 0, y(0) = -5],
  [x(0) = 1, y(0) = 0],
  [x(0) = 2, y(0) = 0],
  [x(0) = 3, y(0) = 0],

```

```

[x(0) = 4, y(0) = 0],
[x(0) = 5, y(0) = 0],
[x(0) = -1, y(0) = 0],
[x(0) = -2, y(0) = 0],
[x(0) = -3, y(0) = 0],
[x(0) = -4, y(0) = 0],
[x(0) = -5, y(0) = 0]

```

```
;
```

```

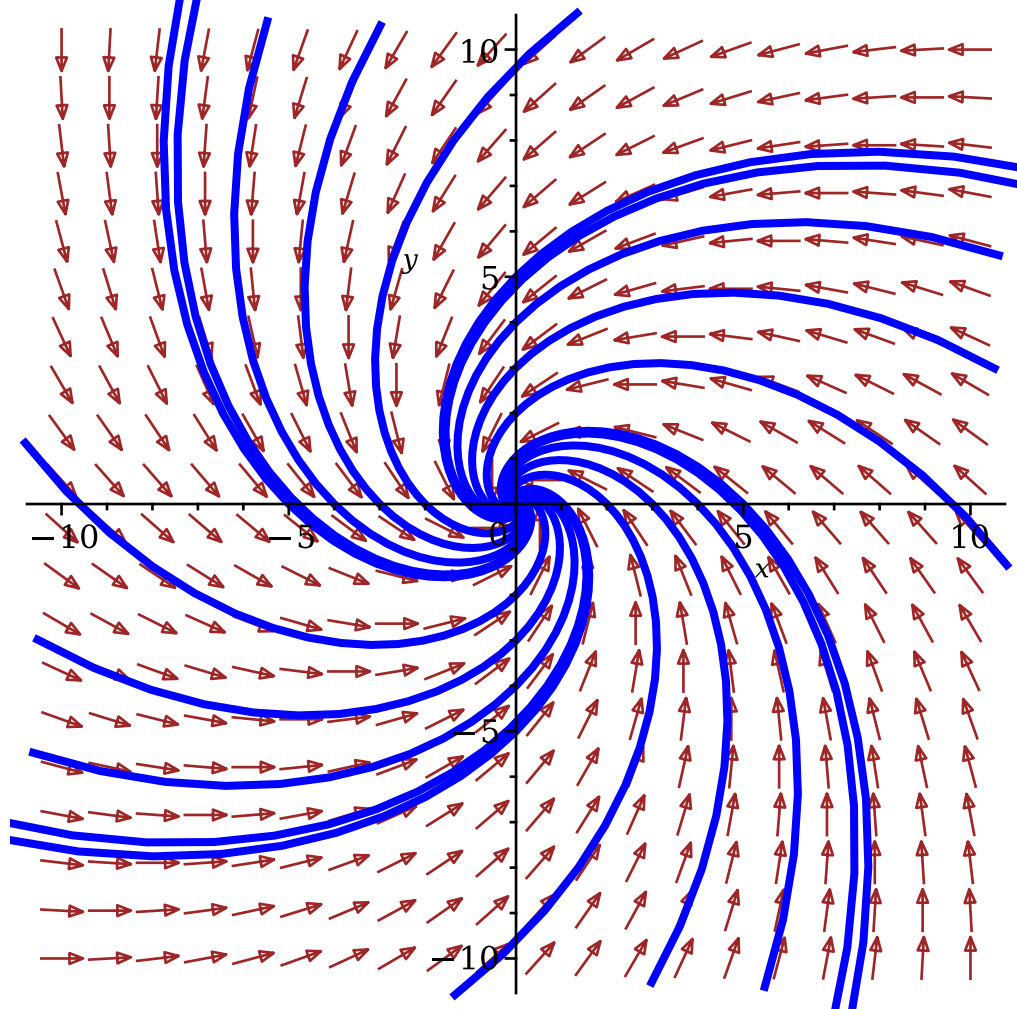
cond_in2 := [x(0) = 0, y(0) = 1], [x(0) = 0, y(0) = 2], [x(0) = 0, y(0) = 3], [x(0) = 0, y(0) = 4], [x(0) = 0, y(0) = 5], [x(0) = 0, y(0) = -1], [x(0) = 0, y(0) = -2], [x(0) = 0, y(0) = -3], [x(0) = 0, y(0) = -4], [x(0) = 0, y(0) = -5], [x(0) = 1, y(0) = 0], [x(0) = 2, y(0) = 0], [x(0) = 3, y(0) = 0], [x(0) = 4, y(0) = 0], [x(0) = 5, y(0) = 0], [x(0) = -1, y(0) = 0], [x(0) = -2, y(0) = 0], [x(0) = -3, y(0) = 0], [x(0) = -4, y(0) = 0], [x(0) = -5, y(0) = 0]

```

```

> DEplot([sist], [x(t), y(t)], t = -5..5, x = -10..10, y = -10..10, [cond_in2],
arrows = medium, linecolor = blue, stepsize = 0.1);

```



```

> restart:
> with(DETools):
> ecdif1 := diff(x(t), t) = -2*x(t):

```

```
> ecdif2 := diff(y(t), t) = -4·x(t) - 2·y(t) :
```

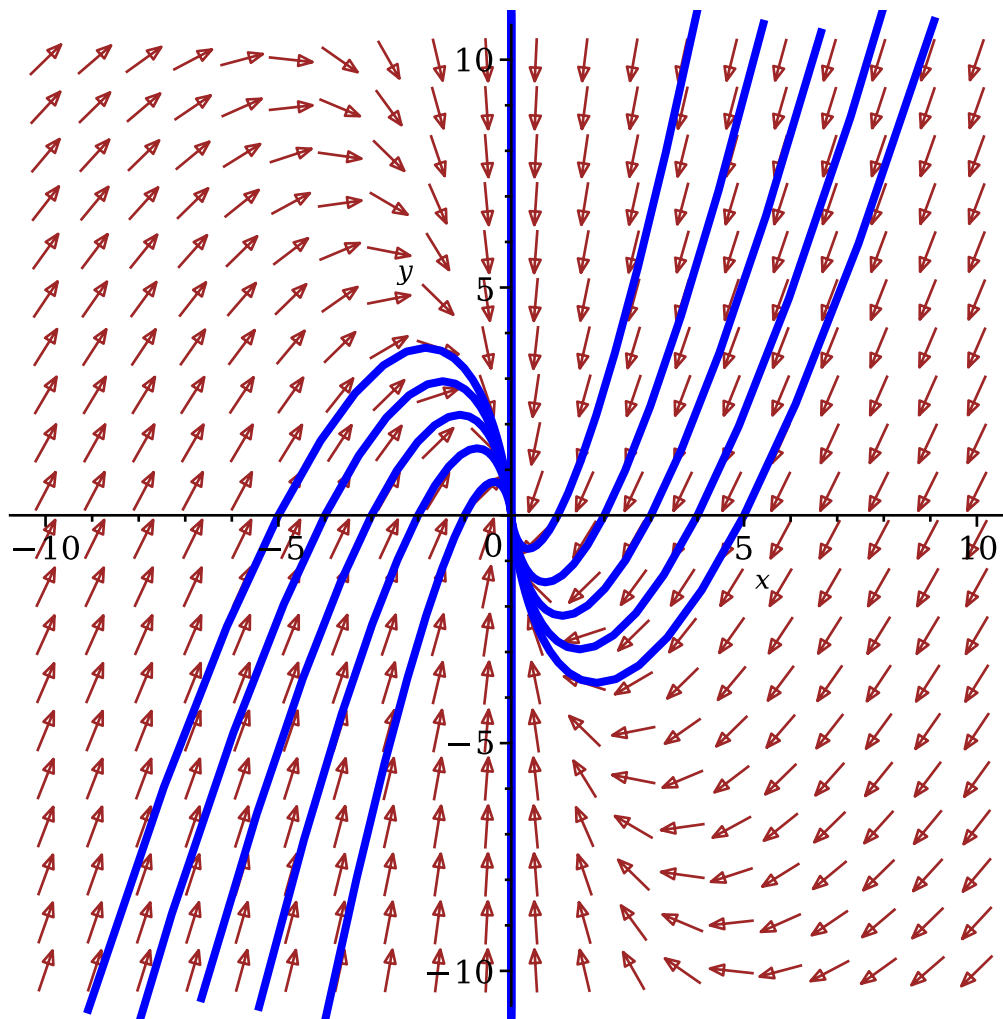
```
> sist := ecdif1, ecdif2 :
```

```
> cond_in2 :=
```

```
[x(0) = 0, y(0) = 1],  
[x(0) = 0, y(0) = 2],  
[x(0) = 0, y(0) = 3],  
[x(0) = 0, y(0) = 4],  
[x(0) = 0, y(0) = 5],  
[x(0) = 0, y(0) = -1],  
[x(0) = 0, y(0) = -2],  
[x(0) = 0, y(0) = -3],  
[x(0) = 0, y(0) = -4],  
[x(0) = 0, y(0) = -5],  
[x(0) = 1, y(0) = 0],  
[x(0) = 2, y(0) = 0],  
[x(0) = 3, y(0) = 0],  
[x(0) = 4, y(0) = 0],  
[x(0) = 5, y(0) = 0],  
[x(0) = -1, y(0) = 0],  
[x(0) = -2, y(0) = 0],  
[x(0) = -3, y(0) = 0],  
[x(0) = -4, y(0) = 0],  
[x(0) = -5, y(0) = 0];
```

```
cond_in2 := [x(0) = 0, y(0) = 1], [x(0) = 0, y(0) = 2], [x(0) = 0, y(0) = 3], [x(0) = 0, y(0) = 4], [x(0) = 0, y(0) = 5], [x(0) = 0, y(0) = -1], [x(0) = 0, y(0) = -2], [x(0) = 0, y(0) = -3], [x(0) = 0, y(0) = -4], [x(0) = 0, y(0) = -5], [x(0) = 1, y(0) = 0], [x(0) = 2, y(0) = 0], [x(0) = 3, y(0) = 0], [x(0) = 4, y(0) = 0], [x(0) = 5, y(0) = 0], [x(0) = -1, y(0) = 0], [x(0) = -2, y(0) = 0], [x(0) = -3, y(0) = 0], [x(0) = -4, y(0) = 0], [x(0) = -5, y(0) = 0]
```

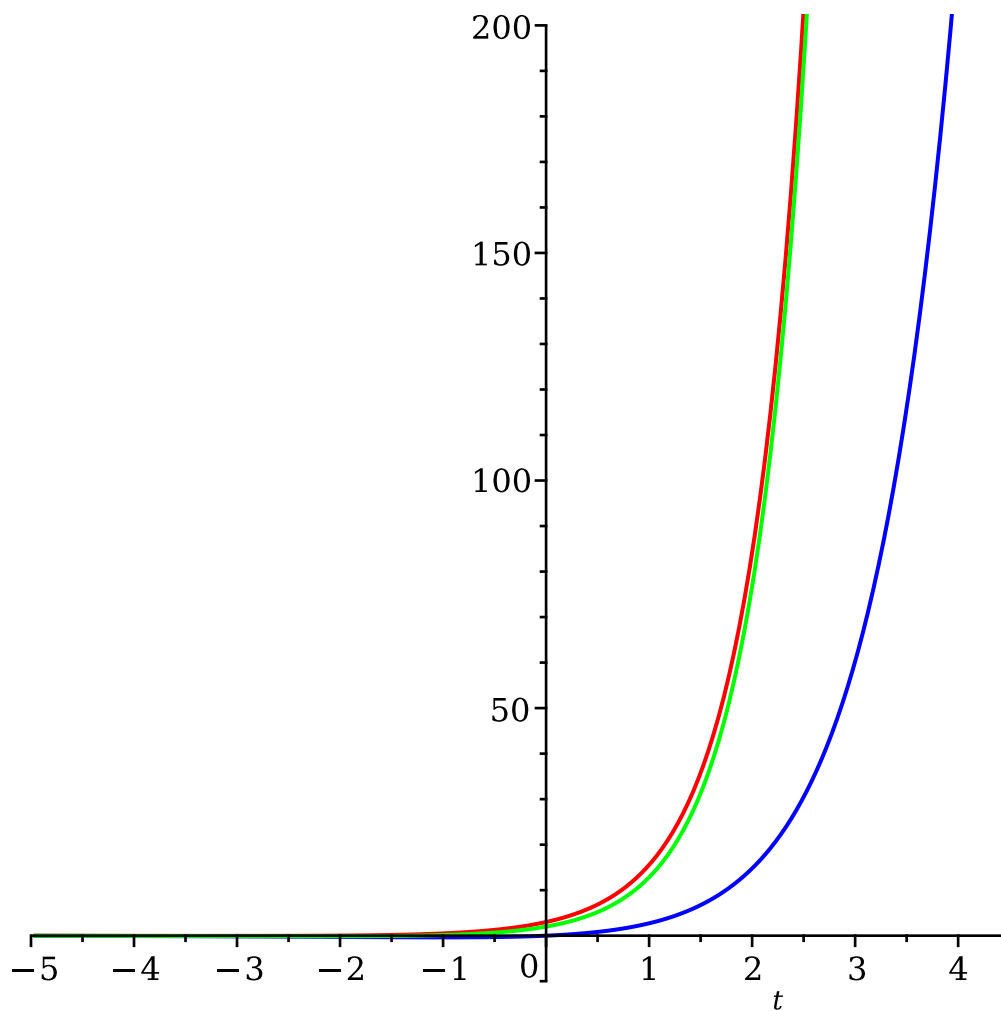
```
> DEplot([sist], [x(t), y(t)], t = -5..5, x = -10..10, y = -10..10, [cond_in2],  
          arrows = medium, linecolor = blue, stepsize = 0.1);
```



```

> restart :
> with(DETools) :
> ecdif1 := diff(x(t), t) = x(t) - y(t) + z(t) :
> ecdif2 := diff(y(t), t) = x(t) + y(t) - z(t) :
> ecdif3 := diff(z(t), t) = -y(t) + 2·z(t) :
> sist := ecdif1, ecdif2, ecdif3 :
> sol := dsolve({sist}, {x(t), y(t), z(t)});
sol := {x(t) = c1 e2t + c2 et + c3 et t + c3 et, y(t) = et (c3 t + c2 - c3), z(t) = c1 e2t
      + c2 et + c3 et t} (16)
> xx := unapply(rhs(sol[1]), t, c1, c2, c3);
      xx := (t, c1, c2, c3) ↦ c1·e2·t + c2·et + c3·et·t + c3·et (17)
> yy := unapply(rhs(sol[2]), t, c1, c2, c3) :
> zz := unapply(rhs(sol[3]), t, c1, c2, c3) :
> plot([xx(t, 1, 1, 1), yy(t, 1, 1, 1), zz(t, 1, 1, 1)], t = -5..5, color = [red, blue,
      green]);

```

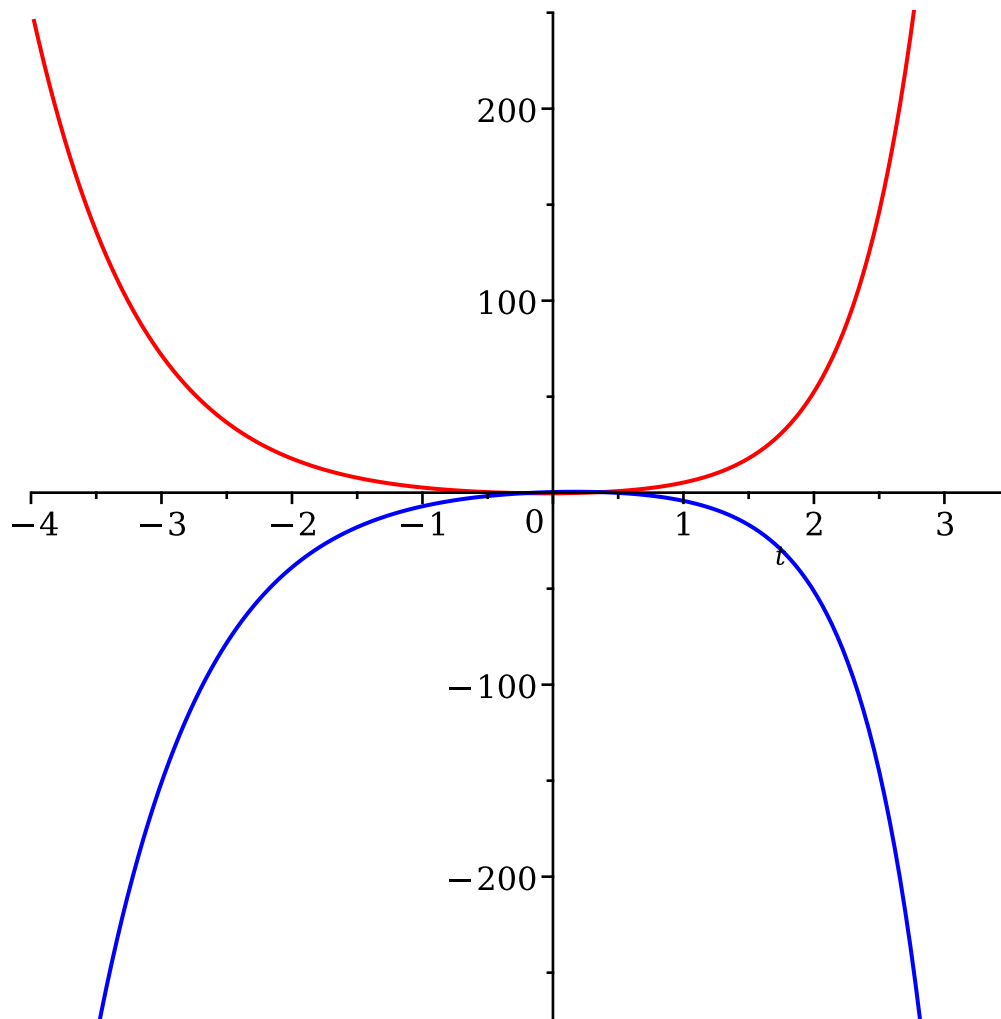


```

> restart:
> with(DETools):
> ecdif1 := diff(x(t), t) = 5·x(t) + 3·y(t) + 1:
> ecdif2 := diff(y(t), t) = -6·x(t) - 4·y(t) + exp(-t):
> sist := ecdif1, ecdif2:
> sol := dsolve({sist}, {x(t), y(t)});
sol := {x(t) = e-t c2 + e2t c1 - 2 - t e-t -  $\frac{e^{-t}}{3}$ , y(t) = -2 e-t c2 - e2t c1 +  $\frac{e^{-t}}{3}$ 
      + 2 t e-t + 3}
> xx := unapply(rhs(sol[1]), t, c1, c2):
> yy := unapply(rhs(sol[2]), t, c1, c2):
> plot([xx(t, 1, 1), yy(t, 1, 1)], t = -4..4, color = [red, blue]);

```

(18)



```

> restart:
> with(DETools):
> ecdif1 := diff(x(t), t) = x(t) + 2·y(t) + 1:
> ecdif2 := diff(y(t), t) = -2·x(t) + 4·y(t) + cos(t):
> cond_in := x(0) = 0, y(0) = 1:
> sist := ecdif1, ecdif2:
> sol := dsolve({sist, cond_in}, {x(t), y(t)}):
> xx := unapply(rhs(sol[1]), t):
> yy := unapply(rhs(sol[2]), t):
> plot([xx(t), yy(t)], t = -5..5, color = [red, blue])

```

